

CHAMBERQ

Hospital & clinic management on Laravel / Filament

Full HMS Build Estimate & Phased Roadmap

How long to build a 13-module Hospital Management System on the ChamberQ stack — measured velocity, bottom-up estimate, and module-by-module delivery plan.

Prepared: 2026-08-09

Status: Planning only — no code changes proposed

Team: Owner + 1 developer + 1 domain/QA

Delivery: Module by module → test → production → next

SECTION 1

Context

ChamberQ is live with real paying clients. The next goal is a full Hospital Management System on the same stack.

Stack: Laravel 12 / Filament 4 / stancl tenancy. Laravel 13 upgrade is explicitly deferred.

Competitor reference

Benchmark bid: **AGMM Soft** → **Moin Uddin Pain Solution, Chattogram**, dated 25 July 2026 — ■380,000 one-time + ■12,000/month, 13 modules, claiming all modules within one month including training.

Delivery model (confirmed)

- **Module by module** — build, test in house, ship to production, then next.
- **Team:** owner + 1 developer + 1 domain/OA person.

Read the one-month claim correctly. AGMM is implementing an existing product sold to Chittagong Medical College, Chevron Clinical Lab, Asian Specialized, and others. Building it took them years. Our comparison is **build-vs-build**.

What the competitor is selling

Every sub-module from the competitor PDF (pages 4–9), mapped against ChamberQ today.

#	Module	Sub-modules	We have today
1	Dashboard & Registration	1	partial
2	Diagnostic Department	47	none
3	Indoor Patient Dept (IPD)	38	~5 (visit records overlap)
4	Outdoor Patient Dept (OPD)	10	~7 — strongest area
5	Emergency	8	none
6	Accounts	13	none — zero money handling
7	Pharmacy	40	none (catalogue only, no stock)
8	Store (Inventory)	24	none
9	HR	~12	none
10	SMS	1	done — prepaid wallet, segment guard
11	Patient & Doctor Portal	1	done
12	MIS	3	~1 (booking KPIs, not financial)
13	Setting & Security	7	~3 (users, roles, organization)
Total		~205	~28 (≈14%)

Velocity baseline

Measured from git log on this repo — not guessed.

Metric	Value
First commit (c8b37ff)	2026-07-25
Latest commit	2026-08-08
Distinct working days	10
Commits	81
app/ LOC	18,206
Blade LOC	9,874
Migrations	45
Models	26
Filament resources	15
Panel pages	49
Services	16
Tests	260

Also: three panels, hybrid path+domain multi-tenancy, production readiness gate, and live paying clients. This is the optimistic bound — greenfield, single-domain, no support load.

SECTION 4

Why HMS work is slower per screen

ChamberQ greenfield velocity does not transfer 1:1 to hospital-grade modules.

1. **Financial correctness** is an invariant problem — ledgers need reconciliation tests, not feature tests.
2. **Domain knowledge** we do not have — referral commissions, BD hospital accounting. Largest schedule risk; domain/QA hire retires it.
3. **Live clients** — HMS migrations touch shared tables; regression and support compete for build hours.
4. **Data migration** from whatever the client runs today.
5. **Module-by-module shipping** adds release, live-DB migration, and pilot-support overhead per module.

The estimate

Bottom-up, weighting ~205 sub-modules by real difficulty.

Weight	Count	Rate	Dev-days
Trivial CRUD setup	~70	5/day	14
Moderate transactional	~85	2/day	43
Heavy engine work	~50	0.5/day	100
Build subtotal			157
Domain discovery & rework (dev share)			15
Data migration + import tooling			10
Hardening, permissions, MySQL validation			15
Bangla staff-panel UI + training (dev share)			6
UAT / pilot fix loop (dev share)			10
Per-module release overhead (13 × ~2d)			26
Total			≈ 239

Two developers parallelize at ~1.7×, not 2× → ≈ **140 working days elapsed**.

Cadence	Elapsed
5 focused days/week	~28 weeks ≈ 6.5 months
4 focused days/week (realistic)	~35 weeks ≈ 8 months

Answer: 7–9 months to all 13 modules with the 3-person team, shipping module by module. First sellable module in **~2–3 months**.

Second method (per-module sums: 47 weeks sequential, ~34 with parallel tracks) lands in the same place.

SECTION 6

Module-by-module roadmap

Build → in-house test → production → next. Ordering by dependency and sellability.

#	Module	Weeks	Why here
1	Settings & Security + Audit Log	2	Prerequisites for every financial screen; audit before money
2	OPD completion — Invoice, Cash Counter, Policy	3	First money on ground; validates billing primitives
3	Diagnostic Department	8	Largest module; biggest seller in BD market
4	Accounts (double-entry core)	6	Needs phases 2–3 generating real money
5	Pharmacy	7	Stock ledger engine — build once here
6	Store (Inventory)	4	Second face on phase 5; parallel with phase 4
7	IPD	9	Heaviest clinical; depends on 3, 4, 5
8	Emergency	2	Thin layer over IPD + billing
9	HR	4	Standalone; parallel with phase 7
10	SMS + Portal extension	1	Extend existing work
11	Unified Dashboard & Registration	1	Built last — unify what exists
Total	Sequential: 47 weeks	~34 w/ parallel	Phases 6 & 9 parallel

SECTION 7

Architectural decisions this plan forces

Expensive to change later. Log in decisions.md when chosen — before code lands.

Ledgers, not mutable columns

Append-only accounting and stock ledgers with balanced-entry invariants at a single write path — same lesson as `GsmText` inside `SmsService::send()`.

Shared-DB multi-tenancy (~150 tables)

Stay shared-DB; invest in tenant-scope enforcement. Per-tenant DB breaks `ResearchDataService` and `SellerOverviewService`.

Bangla staff panel

2026-08-08 made admin English-only. Ward/store/pharmacy staff need Bangla — hundreds of strings if extended like `PatientFacingBanglaTest`.

Real queue worker required

`SendDoctorLateNotices` uses `afterResponse()` because no worker runs. Stock revaluation and period close cannot.

Clinical media backup

Local disk, no off-server backup. Must close before IPD (phase 7), ideally before Diagnostic (phase 3).

Laravel 13

Not upgrading. Write upgrade-clean code over the 8-month build.

Commercial note

Bid: ■380,000 one-time + ■12,000/month $\approx$ \$3,100 + \$100/mo.

7–9 months at that price only works as **SaaS across many centres** — our multi-tenant platform. AGMM appears per-install with denied client DB access.

That asymmetry is the real competitive position — worth more than feature parity.

Verification (reproducible)

```
git log --reverse --format="%ad" --date=short | head -1
git log --all --format="%ad" --date=short | sort -u | wc -l
find app -name "*.php" | xargs wc -l | tail -1
grep -rh "public function test" tests/ | wc -l
```

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